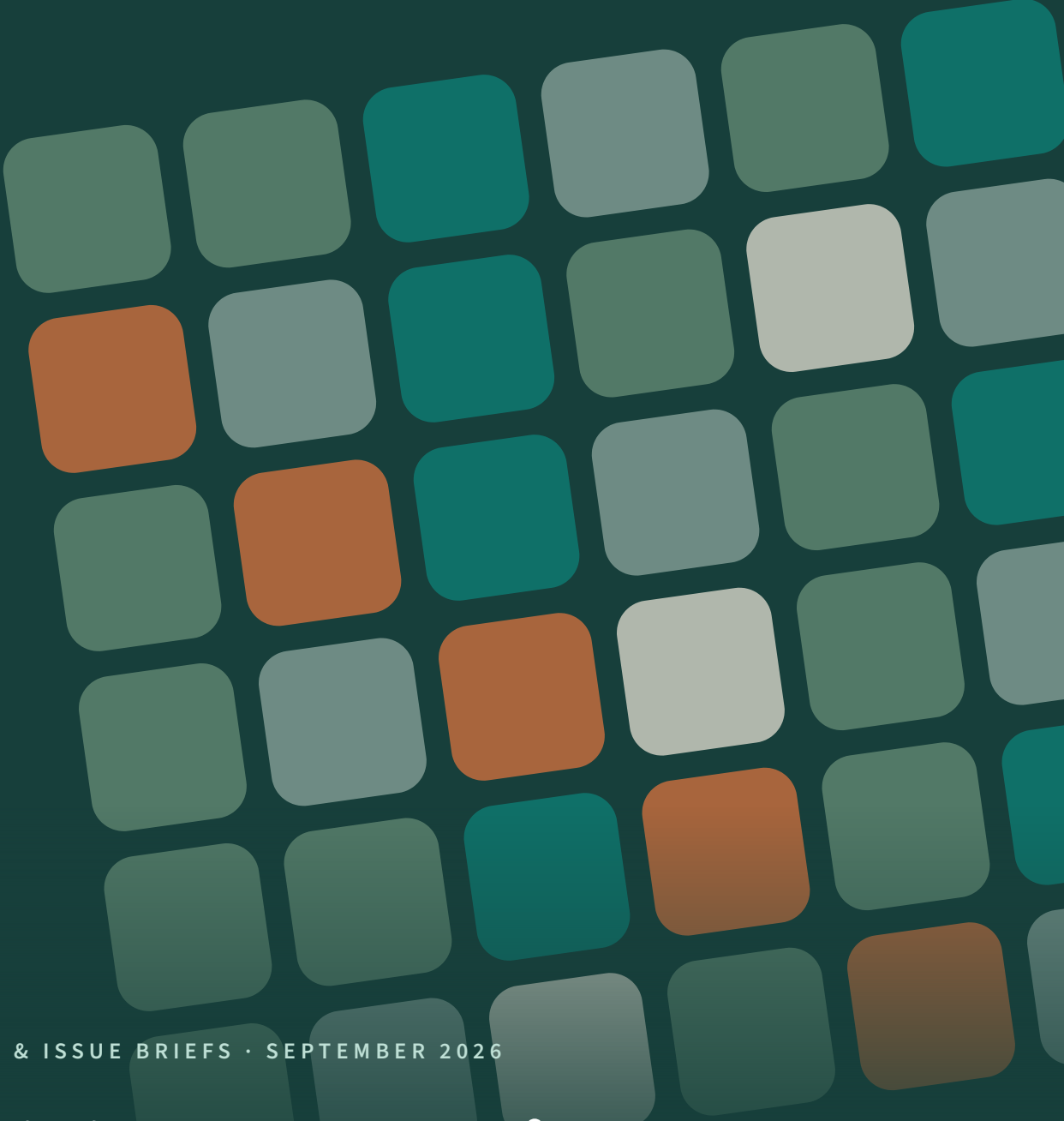


A decorative graphic in the top right corner consisting of a grid of approximately 60 rounded squares. The squares are arranged in a staggered pattern and feature various shades of teal, green, orange, and grey, creating a modern, abstract background element.

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Health Economics Data Finder

How to find evaluation-ready health data, with four briefs
on where health risks cluster, India's position, financial
protection and the health workforce

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Companion to ashuarora.com/data-buff

About this guide

This guide accompanies the **Health Economics Data Finder** (ashuarora.com/data-buff). The Finder is a curated catalogue of 24 evaluation-ready datasets for health economics in India, with global comparators. Part A explains how to use the Finder, how to match data to research designs, and the comparability traps that most often undermine studies. Part B applies the same data to four short issue briefs.

SOURCING

17 of the 24 catalogue entries were checked against official or authoritative sources on 2026-09-29. The rest are marked **To verify**, with the open point stated. The country analysis in Part B is computed from World Bank Group Scorecard vision indicators by a reproducible script. India headline figures in Brief 3 are quoted from official releases and marked **Official release · quoted**; they are not recomputed here.

DISCLAIMER

This is an independent guide by the author. It is not endorsed by any of the data custodians it describes, and it does not host or redistribute their data. Findings and interpretations are the author's own.

SUGGESTED CITATION

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Using the Data Finder

From a research question to a shortlist of datasets, with the traps flagged up front.



A.1 What the Finder does

Each of the 24 datasets is described by the properties that decide whether a study is feasible: its **structure** (panel, repeated cross-section, administrative records, aggregates or modelled estimates), the **finest geography** it supports, its rounds and latest release, whether results are broken down by **sex**, the **research designs** it can serve, its **access** conditions, and its **comparability cautions**. The page has five tools:

Tool	Use it to
Guided search	Rank datasets for a theme, a minimum geography and a design, with the reasons for each match and the main caution.
Catalogue	Search by keyword and filter by scope, theme, access and verification status; open any entry for full details and the official link.
Compare	Sort all datasets side by side by structure, finest level, latest release, sex breakdown and access.
Designs	See which datasets suit each identification strategy.
Hotspot map	See country-level exposure to basic health risks and jump to the datasets that measure them at finer resolution.

A.2 Worked example

“Did a maternal-health programme rolled out district by district increase institutional deliveries?”

Guided search: **Maternal & child health, gender** · **District or finer** · **Difference-in-differences with repeated cross-sections**

1 Health Management Information System Checked

Maternal & child health, gender (main focus) · to village, facility or geocoded household · fits difference-in-differences with repeated cross-sections · open access

Watch out: Counts of services delivered, not population rates; denominators must be estimated.

2 National Family Health Survey Checked

Maternal & child health, gender (main focus) · to district · fits difference-in-differences with repeated cross-sections

Watch out: Infant mortality from NFHS is based on 5-year birth histories and is not directly comparable with annual SRS rates.

3 Demographic and Health Surveys To verify

Maternal & child health, gender (main focus) · to village, facility or geocoded household · fits difference-in-differences with repeated cross-sections · to verify

Watch out: Cluster coordinates are displaced for privacy: up to 2 km (urban), 5 km (rural) and 10 km for 1% of rural clusters.

4 India Human Development Survey Checked

Maternal & child health, gender · to district · fits difference-in-differences with repeated cross-sections

Watch out: Attrition and household splits need handling.

Reading the result. HMIS gives monthly, facility-reported service counts suited to an event study around rollout dates, but it counts services, not population rates. NFHS provides district-representative rates across rounds for a classic difference-in-differences design, but district samples are small. A natural design uses HMIS for timing and dynamics, with NFHS as the population-based check. DHS appears because it fits the design, but India's DHS surveys *are* the NFHS, so it adds cross-country comparison rather than new Indian data.

A.3 Matching data to design

Design	What it needs	Datasets in the Finder
Panel / fixed effects	Follow the same households or people over time; within-unit change removes time-invariant confounders.	Periodic Labour Force Survey, Longitudinal Ageing Study in India, India Human Development Survey, Living Standards Measurement Study
Difference-in-differences with repeated cross-sections	Compare changes across places exposed to a policy at different times; needs consistent geography across rounds.	National Family Health Survey, NSS: Household Social Consumption on Health, Household Consumption Expenditure Survey, Periodic Labour Force Survey, India Human Development Survey, Health Management Information System, Health Dynamics of India, SHRUG: Socioeconomic High-resolution Rural-Urban Geographic Platform, Demographic and Health Surveys, Multiple Indicator Cluster Surveys
Interrupted time series / event study on administrative data	High-frequency records around a rollout date; watch reporting changes and seasonality.	Sample Registration System, Health Management Information System
Spatial / geospatial linkage	Link outcomes to location-based exposures (distance, night lights, climate); respect coordinate displacement.	National Family Health Survey, Census of India 2011, SHRUG: Socioeconomic High-resolution Rural-Urban Geographic Platform, Demographic and Health Surveys, Living Standards Measurement Study
Cross-country comparison	Country-year panels for benchmarking; harmonised definitions but few observations per country.	National Health Accounts Estimates, Demographic and Health Surveys, Multiple Indicator Cluster Surveys, World Development Indicators / Health, Nutrition & Population Statistics, Global Health Expenditure Database, Global Health Observatory, Global Burden of Disease, National Health Workforce Accounts, OECD Health Workforce Migration indicators, Migration & Remittances Data
Descriptive monitoring	Levels and trends for targeting and benchmarking; not causal on its own.	National Family Health Survey, Sample Registration System, NSS: Household Social Consumption on Health, Household Consumption Expenditure Survey, Periodic Labour Force Survey, NSS 64th round: Employment, Unemployment and Migration, Longitudinal Ageing Study in India, Health Management Information System, National Health Accounts Estimates, Health Dynamics of India, NITI Aayog State Health Index, Census of India 2011, Multiple Indicator Cluster Surveys, Living Standards Measurement Study, World Development Indicators / Health, Nutrition & Population Statistics, Global Health Expenditure Database, Global Health Observatory, Global Burden of Disease, National Health Workforce Accounts, OECD Health Workforce Migration indicators, Migration & Remittances Data

A.4 Ten comparability traps

1. **Infant mortality: NFHS vs SRS.** Infant mortality from NFHS is based on 5-year birth histories and is not directly comparable with annual SRS rates.
2. **NSS health rounds.** Recall periods differ (15 days for outpatient care, 365 days for hospitalisation). Questionnaire changes between rounds affect comparability; sub-state estimates are limited to NSS regions.
3. **Consumption surveys.** The 2022–23 and 2023–24 rounds use a revised multi-visit design and are not directly comparable with 2011–12.
4. **PLFS.** The 2025 redesign creates a break in the series. District estimates are available only from 2025.
5. **Migration definitions.** Migration is defined by change in usual place of residence, which differs from the Census definition (place of last residence).
6. **Administrative data (HMIS).** Counts of services delivered, not population rates; denominators must be estimated. Mostly public facilities; reporting completeness varies by state and year.
7. **DHS cluster coordinates.** Cluster coordinates are displaced for privacy: up to 2 km (urban), 5 km (rural) and 10 km for 1% of rural clusters.
8. **Global Burden of Disease.** Modelled, not observed: estimates depend on model assumptions and are revised between rounds.
9. **Health accounts.** Methods are revised between editions; use the latest edition's back-series for trends. National figures (for example India's) may differ from the country's own latest NHA edition.
10. **District boundaries.** District boundaries change between survey rounds and censuses; harmonise to a fixed boundary set (for example 2011) before building district panels.

Where basic health risks cluster

Poverty, food insecurity and missing water, sanitation and hygiene overlap in the same countries.

B1

Question. Are basic health risks spread across many countries, or concentrated in a few that face several at once?

Method. For each country, take the latest value from 2015 onward of five risks: food and nutrition insecurity; the shares of people *without* basic hygiene, sanitation and drinking water (100 minus the reported access share); and poverty at \$3.00 a day. A country is flagged on a risk if it is in the top quarter of countries for that risk (thresholds: food insecurity $\geq 45.5\%$, no hygiene $\geq 73.0\%$, no sanitation $\geq 27.9\%$, no drinking water $\geq 10.8\%$, poverty $\geq 17.5\%$). Only the 80 countries that report all five risks are compared.

Findings. Risks cluster. **31 countries are in the top quarter on at least four of the five risks, and 18 on all five.** 30 of the 31 are in Sub-Saharan Africa; the other is Djibouti. Figure B1.1 shows the 15 countries with the highest burden. Across all countries, poverty moves closely with each risk (Spearman rank correlation with poverty: no hygiene 0.81, $n = 94$; no sanitation 0.79; no drinking water 0.80; food insecurity 0.80; high climate-hazard risk 0.71).

Figure B1.1 15 of the 18 countries in the top quarter on all five risks

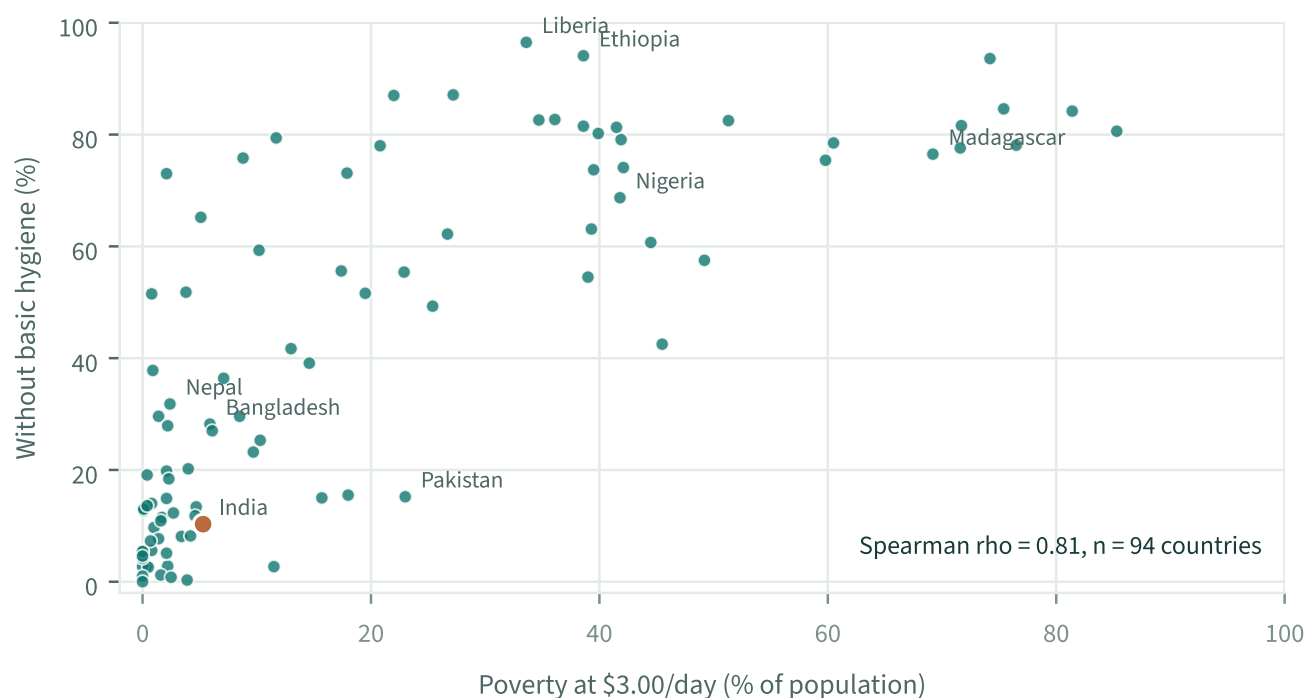
Share of population, latest year from 2015; climate-hazard exposure shown for context (n/a = not reported)

	Food & nutrition insecurity	No basic hygiene	No basic sanitation	No basic drinking water	Poverty (\$3.00/day)	High climate-hazard risk
Liberia	81%	96%	77%	21%	34%	n/a
Ethiopia	61%	94%	90%	44%	39%	31%
Burundi	71%	94%	55%	34%	74%	n/a
Benin	66%	87%	78%	30%	27%	40%
Gambia, The	55%	87%	52%	13%	22%	62%
Malawi	82%	85%	51%	27%	75%	38%
Togo	58%	83%	75%	35%	35%	66%
Tanzania	58%	82%	63%	35%	51%	50%
Sierra Leone	90%	81%	75%	32%	42%	42%
Congo, Democratic Republic of	82%	81%	84%	64%	85%	15%
Guinea-Bissau	62%	80%	71%	38%	40%	n/a
Lesotho	60%	79%	52%	22%	42%	n/a
Niger	51%	78%	84%	47%	60%	69%
South Sudan	88%	78%	86%	60%	76%	n/a
Central African Republic	88%	78%	86%	64%	72%	n/a

Source: author's analysis of World Bank Group Scorecard vision indicators (FY25 cycle); latest country-year from 2015 onward. Ordered by number of risks flagged, then by share without basic hygiene.

Figure B1.2 Poverty and missing basic hygiene go together

Each dot is a country (94 with both measures)



Source: author's analysis of World Bank Group Scorecard vision indicators (FY25 cycle); latest country-year from 2015 onward.

INTERPRETATION AND NEXT STEPS

These are country-level associations, not causal effects: poverty, weak infrastructure and poor services share common causes. The overlap does suggest that single-sector programmes in these countries face compounding risks. To study it within countries, the Finder points to **DHS** and **MICS** (household water, sanitation, hygiene and child nutrition, with displaced GPS clusters in DHS) and to **LSMS** panels for poverty dynamics. A useful design links DHS clusters to climate exposure (spatial linkage), while respecting the coordinate displacement.

India in the global picture

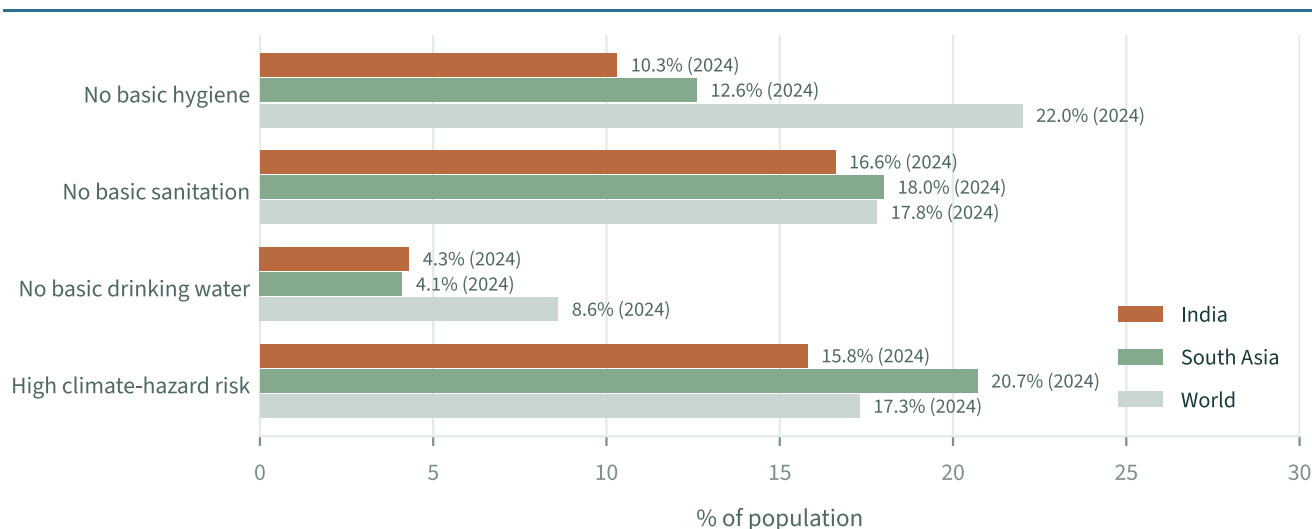
India is below the world average on basic-service deprivation and climate-hazard exposure.

B2

Findings. On the Scorecard's latest values, India fares better than both South Asia and the world on missing basic hygiene (10.3% of the population, against 12.6% in South Asia and 22.0% worldwide) and on high climate-hazard risk (15.8% against 20.7% and 17.3%). It is close to both on sanitation (16.6%) and in line with South Asia on drinking water (4.3%, half the world rate of 8.6%) (Figure B2.1).

Figure B2.1 India, South Asia and the world

Share of population; year of latest value in brackets



Source: author's analysis of World Bank Group Scorecard vision indicators (FY25 cycle); latest country-year from 2015 onward.

TWO CAUTIONS

Food insecurity has no Scorecard value for India, so India cannot be placed on that measure. **Poverty at \$3.00 a day** is reported for India for 2022 (5.3%) but for South Asia and the world for 2024. The years differ and the regional figure includes India, so the two should not be compared directly. National averages also hide large differences between states; the district-level NFHS-6 fact sheets and the Health Dynamics of India tables are the next step for sub-national analysis.

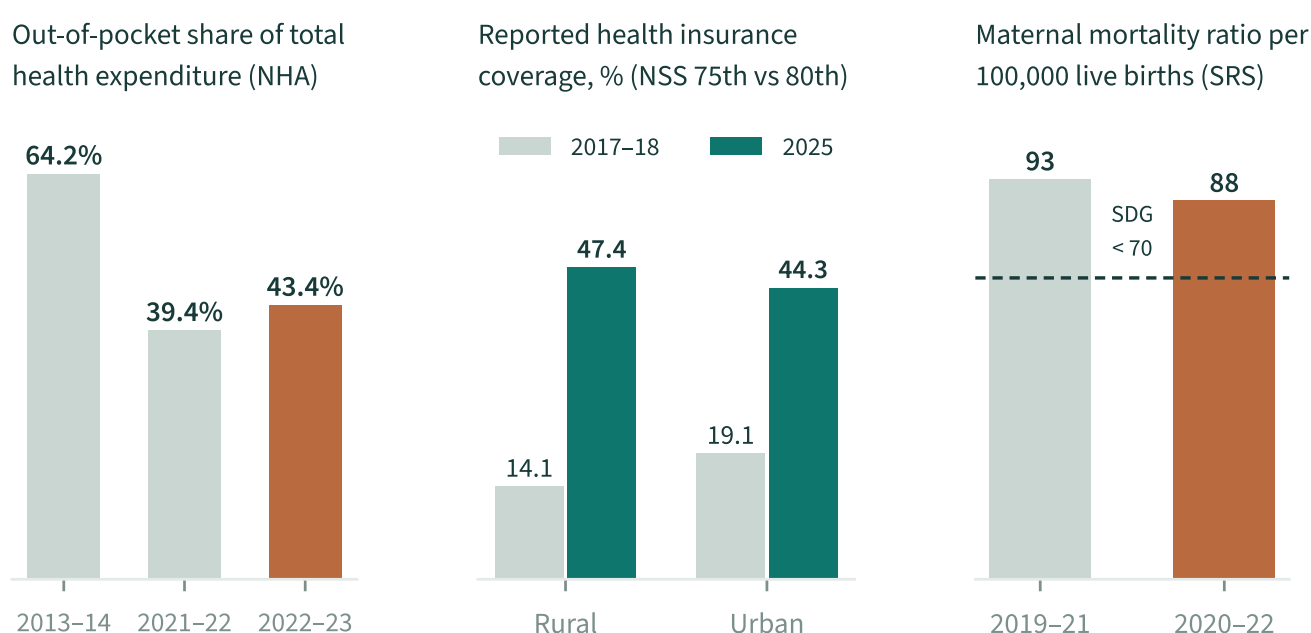
Financial protection and maternal health in India

Insurance coverage has risen sharply; out-of-pocket spending fell for a decade, then ticked up.

This brief quotes headline figures from official releases [Official release · quoted](#). They are reported as published, not recomputed; the Finder entries link to each source.

Figure B3.1 Three official headline indicators

National Health Accounts; NSS Household Social Consumption: Health; Sample Registration System



Sources: NHA Estimates for India 2022-23 (NHSRC/MoHFW); NSS 80th round, Household Social Consumption: Health, January-December 2025, compared with the 75th round (2017-18) (NSO, MoSPI); SRS Special Bulletin on Maternal Mortality 2020-22 (Office of the Registrar General). As reported in the official releases.

Financial protection. Out-of-pocket spending fell from 64.2% of total health expenditure in 2013-14 to 39.4% in 2021-22, then rose to 43.4% in 2022-23. Over roughly the same period, reported health insurance coverage rose from 14.1% (rural) and 19.1% (urban) in 2017-18 to 47.4% and 44.3% in 2025, with rural coverage now above urban.

Maternal health. The maternal mortality ratio fell from 93 per 100,000 live births in 2019-21 to 88 in 2020-22. The SDG 2030 target is below 70.

RESEARCHABLE QUESTIONS AND DATA

- **Did the insurance expansion reduce catastrophic health spending?** Compare NSS 75th (2017-18) and 80th (2025) rounds across states with different scheme roll-out intensity (difference-in-differences with repeated cross-sections). Watch the recall periods and questionnaire changes, and the 2022-23 rise in out-of-pocket share, which suggests coverage has not translated evenly into protection.
- **Is the maternal-mortality gap now about quality rather than access?** With institutional delivery near universal, use HMIS for service-quality indicators over time (interrupted time series), NFHS-6 for antenatal-care content by district, and SRS for mortality trends. SRS reports maternal mortality only for larger states, in three-year pools.

Health workforce and migration

B4

India is a major origin of doctors and nurses working in OECD countries, yet domestic workforce data sit in separate sources.

What is known. The OECD's *International Migration Outlook 2025* identifies India among the main countries of origin of both foreign-born doctors and nurses working in OECD countries. At home, the Health Dynamics of India tables report sanctioned posts, staff in position and shortfalls in public facilities by district.

The measurement gap. No single source links training, domestic deployment and international migration of health workers. The Finder's migration and workforce sources each cover one piece:

Question	Datasets
How many health workers are in post, where, and what are the shortfalls?	Health Dynamics of India (district tables); WHO National Health Workforce Accounts (cross-country density)
Who works in health occupations, and on what terms?	PLFS (occupation, wages; district-representative from 2025)
How many leave, and to where?	OECD health workforce migration indicators (destination-country data); KNOMAD migrant stocks and remittances
Internal migration patterns	NSS 64th round (2007–08), PLFS 2020–21 migration module, Census 2011 D-series tables

RESEARCHABLE QUESTION

Does international recruitment of nurses reduce nurse availability in origin states? One design combines destination-country demand shocks (from OECD inflow data) with state-level exposure through historical migration networks (a shift-share design), and measures domestic outcomes with Health Dynamics of India staffing and PLFS employment. Key cautions: OECD data count workers by country of training or birth, not Indian state of origin, and public-sector staffing tables miss the private sector.

Limitations, catalogue and references

What this guide can and cannot support.

- **Catalogue.** Metadata only. 7 entries are still marked to verify. Details such as release status change; check the custodian's site before starting a study.
- **Country analysis.** Country values use each country's latest year from 2015 onward, so years differ across countries. The hotspot comparison covers only the 80 countries reporting all five risks, which over-represents lower-income countries. Correlations are descriptive.
- **Official figures.** Brief 3 quotes published headline figures; definitions (for example, whether coverage is counted per person or per household) follow the original releases.

Table C.1 The catalogue

Dataset	Scope	Structure	Finest level	Access	Status
National Family Health Survey	India	Repeated cross-section (household, women, men, children)	District	registration	Checked
Sample Registration System	India	Aggregate vital rates from continuous registration in sample units	State	open	Checked
NSS: Household Social Consumption on Health	India	Repeated cross-section, unit-level (household, person)	NSS region (sub-state)	registration	Checked
Household Consumption Expenditure Survey	India	Cross-section, unit-level (household)	NSS region (sub-state)	registration	Checked
Periodic Labour Force Survey	India	Repeated cross-section with an urban rotational panel	District	registration	Checked
NSS 64th round: Employment, Unemployment and Migration	India	Cross-section, unit-level (household, person)	NSS region (sub-state)	registration	Checked
Longitudinal Ageing Study in India	India	Panel (adults aged 45+ and spouses)	State	registration	To verify
India Human Development Survey	India	Panel (households)	District	registration	Checked
Health Management Information System	India	Administrative, monthly facility reporting	Village, facility or geocoded household	open	Checked
National Health Accounts Estimates	India	Aggregate annual accounts	Country	open	Checked
Health Dynamics of India	India	Administrative, annual	District	open	Checked
NITI Aayog State Health Index	India	Composite index	State	open	To verify
Census of India 2011	India	Census tables	Village, facility or geocoded household	open	To verify

SHRUG: Socioeconomic High-resolution Rural-Urban Geographic Platform	India	Linked geospatial data across sources	Village, facility or geocoded household	open	Checked
Demographic and Health Surveys	Global	Repeated cross-section	Village, facility or geocoded household	registration	To verify
Multiple Indicator Cluster Surveys	Global	Repeated cross-section	NSS region (sub-state)	registration	Checked
Living Standards Measurement Study	Global	Cross-sections and household panels	Village, facility or geocoded household	registration	To verify
World Development Indicators / Health, Nutrition & Population Statistics	Global	Country-year indicators	Country	open	To verify
Global Health Expenditure Database	Global	Country-year health accounts	Country	open	Checked
Global Health Observatory	Global	Country-year indicators	Country	open	To verify
Global Burden of Disease	Global	Modelled estimates	State	open	Checked
National Health Workforce Accounts	Global	Country-year indicators	Country	open	Checked
OECD Health Workforce Migration indicators	Global	Country-year, destination and origin	Country	open	Checked
Migration & Remittances Data	Global	Country and country-pair (bilateral) matrices	Country	open	Checked

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6. Dataset custodians and sources for each catalogue entry are linked in the Finder (ashuarora.com/data-buff).